nanoker

CERAMIC TECHNICAL DATA SHEET

KTM650

Carbon based Composite - Heat Spreader

KEY PROPERTIES

- Extremely high thermal conductivity
- Low density
- Excellent machinability
- CTE matching with silicon
- Suitable for complex geometries

APPLICATIONS

- Diode lasers
- Power transistor substrates
- LED substrates
- Wireless communications
- Electrical drives
- Aerospace power supplies
- Medical electronic systems
- Avionics systems

MAIN PROPERTIES

Property	Value	Unit
Density	2.57	g/cm³
Flexural strength (XY/Z)	102 / 16.9	MPa
Young Modulus (XY/Z)	69.7 / 5.5	GPa
Thermal Conductivity (XY)	650	W/m·K
Thermal Conductivity (Z)	45	W/m·K
Thermal Diffusivity (XY)	390	mm²/s
Thermal Diffusivity (Z)	27	mm²/s
CTE average	6.5	$x10^{-6}$ K^{-1}
CTE (XY/Z)	2.4 / 14.7	$x10^{-6}$ K^{-1}
Specific Heat	0.65	J/g·K
Electrical Conductivity	0.8	MS/m

** All properties measured at 20ºC unless otherwise stated.*
*** Measured in a wall thickness of 2 mm*

OPERATING CONDITIONS

Max working temp (air)	400	ºC
Recomended temp	<350	ºC
Continuous temp	300	ºC

ELECTRICAL BEHAVIOR

Conductive

Electrical Conductivity	0.8 MS/m

MATERIAL INFORMATION

Material type	Carbon-based composite
Structure	Anisotropic composite
Key feature	Thermal management
Machinability	Excellent
Color	Black

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

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